Service Design

Service Design

- Service design is involved in planning both new services and changes to existing services, ensuring that the new or changed service fulfills the service strategy by delivering the business objectives
- It touches all areas of IT because each area will have a role in delivering and supporting the eventual service
- It includes the design of both services and the service management processes, which will ensure that the service continues to provide value
- Successful design depends on taking the time to plan ahead
- This is often seen as wasted time, with the service provider anxious to press ahead with the design
- In fact, nothing could be further from the truth; insufficient planning leads to costly rework and delay late
- Part of the planning phase is identifying and managing risks to ensure a successful outcome

Purpose of Service Design

- The purpose of service design is to deliver a new service or a change to an existing service that is capable of delivering the strategic outcome required
- This involves not only the technology used to deliver the service but the processes and policies needed to ensure that the technical solution delivers the intended value
- It considers what will be required by the transition phase to implement the service in the operational environment, how the service will perform, and what will be required to support it
- Service design must ensure that the service runs within the allocated budget and delivers a level of service that meets or exceeds the customer requirement

The Objectives of Service Design

- In an ideal world, services would be designed so well that they would require little or no improvement after their deployment
- Although this is not realistic, service design should aim to deliver a service that will require very little improvement later
- This is not to say that there is no role for continual service improvement; every design effort should apply lessons learned from previous design projects because these lessons can improve the eventual output
- The business requirement may also change, requiring improvements to the original design
- Service design activities may be undertaken on a regular basis or as the result of identifying a new or changed business requirement

The Scope of Service Design

- ITIL service design considers not only the current requirement; it extrapolates from this to identify possible future needs and ensures that the service being designed will be able to be developed to meet these requirements too
- It ensures that the design fits the requirements, and tries to take advantage of technical developments to deliver an innovative service
- It is essential that the service is aligned with the business need, so service design describes how to identify these requirements and ensure that the service delivers what the customer requires
- It considers the functional requirements that must be delivered as well as the required service levels
- It ensures that any design constraints are identified and adhered to

The Scope of Service Design

- The processes included within service design are as follows:
  - Design coordination
  - Service catalog management
  - Service level management
  - Availability management Capacity management
  - IT service continuity management
  - Information security management
  - Supplier management
- All processes across the lifecycle must be linked; it is important to understand how these various processes interact so that an amendment to one process does not have unintended consequences on another

The Value Service Design Provides to the Business

- As we mentioned, good service design delivers high-performing services that match the business requirement and are delivered at an affordable cost. Following the service design guidelines in the ITIL framework will have a number of positive outcomes for the business.
- Foremost among these is a lower total cost of ownership across the lifetime of the service, because all aspects of the design, processes, and technology have been considered and designed to work together, therefore minimizing later rework
- Good design will also consider the corporate strategy and existing architecture, as well as any design constraints, to ensure that the service operates within these parameters
- By ensuring that the warranty aspects of the service are included during the design stages, the service design delivers a reliable, effective service that meets the customer requirement
- The service will be designed to meet the agreed service level requirements and will ensure the required capacity, availability, and service continuity requirements are met cost-effectively

The Value Service Design Provides to the Business

- Service design considers not only how the service will perform in the live operational environment; it also considers the most effective way of enabling the transition from design to live, ensuring that all the required information for this transition stage is captured in a service design package (SDP)
- Service design will consider what metrics and controls may be required for good governance and ensure that these are part of the design
- Ensuring that the design gathers the appropriate metrics will assist in the continual service improvement of the service in the future
- It will also consider what processes are required and ensure that these are designed to be both effective and efficient
- Finally, service design will consider particular strategic requirements, such as using cloud technologies or delivering services with the lowest possible carbon footprint, ensuring that the design is aligned with these business requirements

Describing the Service

- ITIL recommends producing key output from the service design stage: service design package (SDP)
- The service design package consists of one or more documents, produced during the service design stage, that describe all aspects of the service, throughout its lifecycle
- Typical contents of an SDP include the following:
  - Original agreed business requirements for the service
  - How the service will be used
  - Key contacts and stakeholders
  - Functional requirements
  - Management requirements
  - Service level requirements
  - Technical design of the new or changed service including hardware, software, networks, environments, data, applications, technology, tools, and documentation
  - Sourcing strategy
  - New or changed processes required to support the service
  - Organizational readiness assessment

Key Elements of Service Design

Key Elements of Service Design

- Service design, as described within the ITIL framework, takes a holistic view of what is required to design and deliver a service
- A frequent error in service design is that this view, encompassing all aspects, is replaced with a narrow, technical view of the design
- Such a design may work well in theory but fail to deliver in practice because of the lack of attention given to changes to processes that may be required, the skills required for users and those providing support of the service, or the poor choice of third-party suppliers
- Successful service design must therefore consider the four key elements, sometimes known as the four p's:
  - People
  - Processes
  - Products
  - Partners

People

- The best technical design will fail if the people who need to use it or support it are not adequately prepared
- The people element of service design ensures that the human aspect is not forgotten
- Service designers must consider how many people will be required to support the new service (resource) and what skill set they will require to do so effectively (capability)
- Training in the specific processes required to support the service will also be necessary
- The people who will be using the service will also need training in its use in order to gain the full benefit of the service
- If existing staff members are to be used, a training needs analysis will be required, and adequate time and money need to be budgeted for training
- A communications plan that ensures the right information is given to the right people at the right time by the most appropriate method will also be needed in order to ensure that the staff members understand what is required of them

Processes

- The second element to be considered is that of processes
- In addition to the service design processes described in the ITIL framework, the new service may require additional processes to be designed, such as an authorization or procurement process
- Some IT service providers may ignore the need to design either of these groups of processes in order to shorten the “speed to market” time; this is a false economy
- Failure to consider the future capacity requirements of the service—for example, because the design appears to be adequate—could cause problems later when the service is unable to support increased demand without a major redesign
- If the new service becomes business-critical, failure to design sufficient resilience may mean that poor availability impacts the business
- As part of service design, processes should be documented, together with the interfaces between them and other processes
- All existing processes across the lifecycle should be assessed to identify whether any changes to them are required to enable this new or changed service to become operational and be supportable

Products

- Products are not only the services that result from the service design stage itself but also the technology and tools that are chosen to assist in the design or to support the service later
- So, for example, the service design may be for an online shopping service, and the other products may include a credit-card processing application, an automatic stock reordering service when stock levels reach a threshold, monitoring tools to alert the service provider if user response time exceeds a set time, and so on

Partners

- Partners are those specialist suppliers—usually external third-party suppliers, manufacturers, and vendors—that provide part of the overall service
- Ensuring the correct supplier is chosen is essential, because failure by a supplier may cause the IT service provider to breach an agreed service level
- External suppliers are managed through the supplier management process, which ensures that the necessary contracts are put in place and monitors the delivery by the supplier against the contract terms

Aspects of Service Design

Major Aspects of Service Design

- Concentrating on just the service solution will not be sufficient; other aspects need to be considered
- The design process must take a holistic approach to designing new or changed services, because success depends not only on the technical solution but also the management and architectural environment in which that solution is to operate, the processes and skills that will be required to ensure it runs effectively, and the metrics that need to be provided to monitor and manage it
- There are five aspects of service design
  - Service solutions
  - Tools and systems for management information
  - Architectures
  - Measurement systems
  - Processes

Service Solutions

- The first aspect is the solution itself; in other words, the new functionality offered by a new application or other service
- The requirements will have been defined in the service portfolio; it is the job of service design to deliver a solution that meets these requirements within any technical or financial constraints that exist
- The solution must conform to corporate rules and must work with the existing services
- The supporting services must be able to deliver to the level required; if this is not possible, the overall design will need to be altered
- It is likely that a structured design approach will be used to deliver the service solution, ensuring that it meets the required utility, warranty, and service levels and that it is delivered on time and within budget
- As the service lifecycle progresses, requirements may change, so the approach must be not only structured but also flexible enough to meet the developing requirements

Management Information Systems and Tools

- Management information tools and systems are used to support and automate processes
- They are usually part of a bigger framework of policies, processes, functions, standards, guidelines, and tools in use within the organizations
- Organizations may have (for example) a quality management system, an information security management system, and the service management system
- The second aspect to be considered when designing a service, therefore, concerns ensuring that the new or changed service will integrate with existing management information systems, such as the service knowledge information system (SKMS)
- The SKMS holds details about the service within the service portfolio and service catalog
- These management information systems need to be examined to ensure that they are able to provide the information required to manage the service effectively
- These management information systems must also be compatible with the rest of the management information framework in use within the organization, such as its quality and security management systems

Architectures

- The organization will have existing systems, and the new or changed service will need to be compatible with them
- This is another aspect to be considered during design
- There will be an existing architectural platform as well as agreed technical standards, and they will need to be evaluated to see whether they can support the new service
- If not, then either the architectures or management systems will need to be amended or the design of the new service will need to be revised

Measurement Systems

- The next aspect of service design is that of measurement
- The design needs to ensure that what is being measured meets the requirement
- The current metrics that are gathered must be sufficient to enable the service to be assessed for efficiency and effectiveness
- If not, then the measurement methods will need to be improved, or the service metric requirement will need to be altered

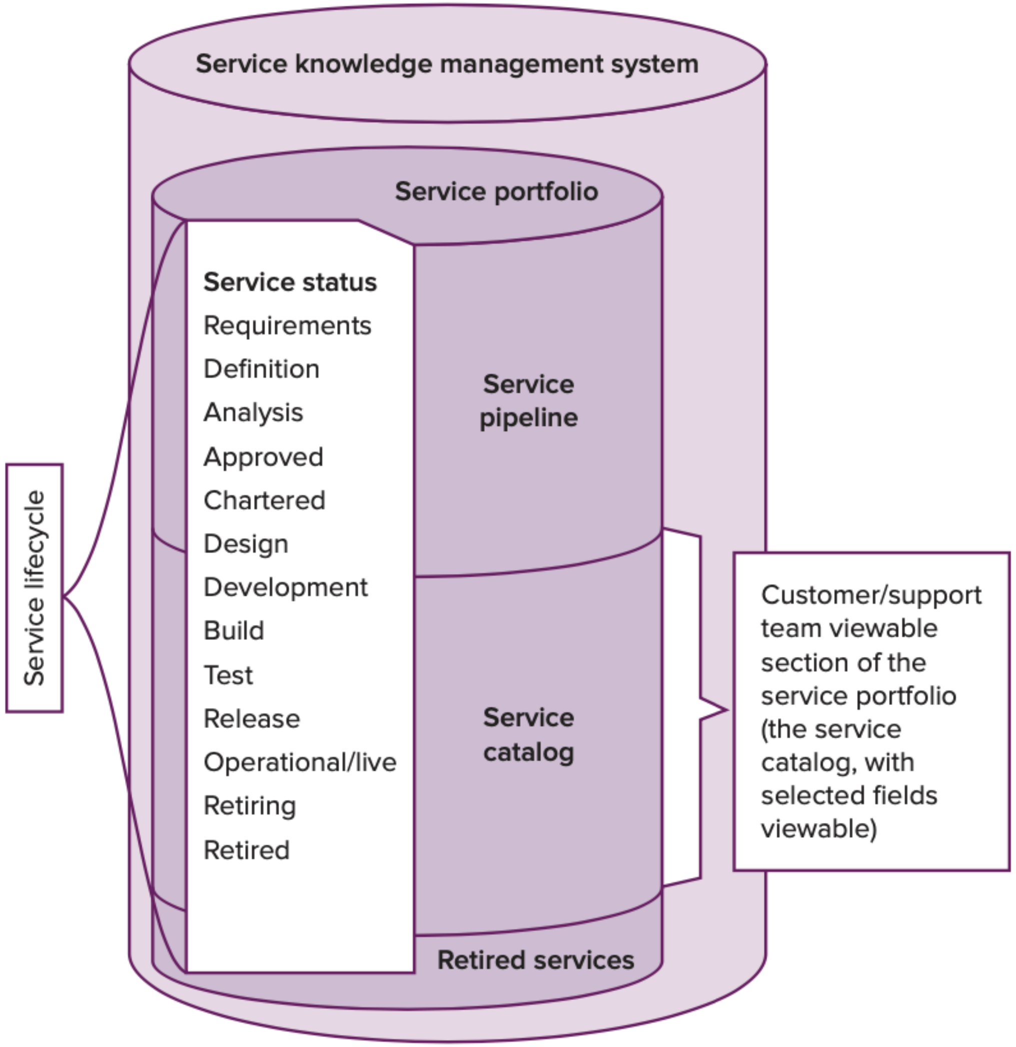
Processes

- Each new or changed service will require processes
- These processes may be existing processes or specific to the service
- Any processes, roles, responsibilities, and skills that will be required by the new or changed service need to be considered
- New processes will need to be designed, and existing processes need to be checked to see whether they require any improvement in order to support the new service
- If they cannot be improved, the design of the service will need to be amended
- Be aware that the processes being evaluated here are not only the processes under service design, but all IT and service management processes throughout the lifecycle
- Process models can be used to clarify the processes

Service Design Processes

Service Catalog Management

- A database or structured document with information about all live IT services, including those available for deployment
- The service catalogue is part of the service portfolio and contains information about two types of IT service:
  - customer-facing services that are visible to the business and
  - supporting services required by the service provider to deliver customer- facing services.



Availability Management

- The availability of a service is critical to its value
- No matter how clever it is or what functionality it offers (its utility), the service is of no value to the customer unless it delivers the warranty expected
- Poor availability is a primary cause of customer dissatisfaction
- Availability is one of the four warranty aspects that must be delivered if the service is to be fit for use
- Targets for availability are often included in service level agreements, so the IT service provider must understand the factors to be considered when seeking to meet or exceed the availability target

Calculating Availability

- **Example A**:
  - A service is available 24 hours a day, 7 days a week
  - One hour of downtime per week is calculated as follows:
    - 168 hours – 1 hour downtime = 167/168 × 100 = 99.4% availability
- **Example B**:
  - If the service is available but used only 9 a.m. to 5 p.m., Monday to Friday (and these 40 hours are the service hours agreed in the SLA), then the same one hour of downtime results in a different figure:
    - 40 hours – 1 hour downtime = 39/40 × 100 = 97.5% availability

- If the downtime occurred overnight, it would be included in the calculations in Example A but not those in Example B, because there was no agreed service after 5 p.m.
- It is important therefore to agree on exactly what the agreed service hours are; they should be documented in the SLA
- The basis for the calculation should be clear to the customer.

MTRS

- When a fault occurs and there is insufficient resilience in the design to prevent it from affecting the service, the length of the downtime that results can be affected by how quickly the fault can be overcome
- This is called maintainability and is measured as the mean time to restore service (MTRS)
- It may be more cost-effective to concentrate resilience measures for those items that have a long service restoration time
- To calculate MTRS, divide the total downtime by the total number of failures
- A service suffers four failures in a month. The duration of each was 1 hour, 2 hours, 1.2 hours, and 1.8 hours, resulting in a total downtime of 6 hours.
  - Availability Management 109
  - MTRS = 6 ÷ 4 = 1.5 hours

Capacity Management

- One of the responsibilities of the service provider is to ensure that the service is able to cope with the demands put upon it
- If the design is not able to cope, the service will fail to meet the performance specified in the SLA
- In addition, the service must be designed to meet not only the requirements for today; it must anticipate future requirements and be able to meet them, too
- Overspecifying the design would ensure that it could meet future demands; however, this may result in wasted expenditure if the actual demand does not come close to the capabilities of the design
- ITIL states that capacity management is responsible for ensuring that the capacity of IT services and the IT infrastructure is able to meet agreed current and future capacity and performance needs in a cost-effective and timely manner
- The capacity management process has therefore to understand the likely changes in capacity requirements and ensure that the design and ongoing management of the service meet this demand
- Delivering sufficient capacity is a key warranty aspect of a service that needs to be delivered if the benefits of the service are to be realized